CancerOS — Product Requirements Document

Project: CancerOS — AI-Powered Breast Cancer Digital Twin
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Product type: Multimodal AI research and simulation platform
Initial scope: Breast cancer only
Primary users: Oncology researchers and clinicians acting in a research capacity
Development objective: Working hackathon proof of concept
Strategic objective: Build a technically competitive prototype suitable for consideration as a hackathon finalist

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1. Executive Summary

CancerOS is an AI-powered breast cancer digital twin research prototype that combines authentic medical imaging, available clinical observations, simulated longitudinal patient records, tumor reconstruction, and multimodal AI interpretation to investigate how breast tumors and patient conditions change over time.

The prototype addresses a fundamental limitation of isolated clinical observations: a single scan or laboratory report provides an incomplete representation of a disease that evolves over time.

CancerOS transforms disconnected observations into a patient-specific, time-indexed computational representation.

The system shall allow researchers to:

1. Import real breast cancer imaging and available clinical metadata.
2. Construct an anatomical digital twin from imaging-derived tumor segmentation.
3. Generate alternative synthetic longitudinal patient trajectories from a real baseline.
4. Represent different tumor responses, treatment exposures, toxicity patterns, and clinical outcomes.
5. Synchronize imaging changes with blood tests, vital signs, symptoms, and recorded treatment events.
6. Explore patient evolution through an interactive 3D visualization and temporal dashboard.

7. Add new observations and update the digital twin without changing historical states.
8. Generate evidence-linked AI interpretations that distinguish measured findings from simulated or predicted findings.
9. Retrieve relevant historical clinical observations or published research evidence.
10. Evaluate selected capabilities against genuinely observed, held-out patient data wherever available.

The system is intended for research, education, and demonstration, not clinical treatment selection.

1.1 Product definition

CancerOS is a **longitudinal multimodal cancer research and simulation platform** with an interactive patient-specific digital twin.

It is not:

- An electronic health records platform.
- A hospital management system.
- An autonomous diagnostic system.
- A clinically validated cancer treatment predictor.
- A prescription or treatment optimization engine.
- A whole-body physiological simulator.

1.2 Central product proposition

CancerOS transforms a real breast cancer baseline into an evolving digital twin that researchers can observe, update, compare, and analyze across actual and hypothetical disease trajectories.

The digital twin must be more than an animated tumor.

It must include anatomically grounded findings, time-indexed clinical observations, treatment exposures, patient-condition indicators, data provenance, and interpretable AI-generated research observations.

1.3 Primary product demonstration

The principal demonstration shall use a real de-identified breast MRI baseline and five alternative synthetic clinical scenario families.

The researcher shall be able to select one scenario, navigate its follow-up visits, observe the corresponding tumor geometry and clinical measurements, and request an AI interpretation.

A separate demonstration shall use genuinely repeated imaging observations from an available longitudinal dataset to establish that the prototype can represent actual tumor changes rather than synthetic trajectories alone.

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2. Product Goals and Success Measures

2.1 Primary goals

G-01 — Functional digital twin

Demonstrate construction of a patient-specific breast-region tumor visualization from real source imaging or an associated verified segmentation mask.

G-02 — Longitudinal intelligence

Demonstrate that the system can represent and compare successive patient states, including tumor measurements, laboratory observations, vital signs, symptoms, and treatment events.

G-03 — Synthetic clinical simulation

Build a reproducible simulation system that generates coherent longitudinal research records from authentic baseline observations.

G-04 — Multimodal AI interpretation

Demonstrate structured AI-assisted interpretation that considers more than one clinical information type and cites the underlying evidence.

G-05 — Dynamic state updates

Allow subsequent observations to be submitted without overwriting prior states, and recompute relevant comparisons and insights.

G-06 — Scientific transparency

Differentiate observed, derived, synthetic, interpolated, and predicted data throughout the application.

G-07 — Technical credibility

Provide measurable results, independently testable functionality, and evidence of performance on relevant held-out data.

G-08 — Hackathon competitiveness

Deliver an integrated, visually compelling, reliable demonstration with clear scientific novelty and realistic future applications.

2.2 Success targets

Metric	Prototype target
Cancer types supported	One: breast cancer
Initial baseline cohort	Up to 100 distinct usable cases
Synthetic scenario families	Five
Planned synthetic branches	Up to 500
Minimum observations per demo branch	Four dated visits
Planned synthetic visit records	Up to 2,000
Curated interactive demonstration cases	At least five
Authentic repeated-imaging demonstration	At least one suitable real longitudinal case
Imaging visualization	Rotatable breast-region tumor representation
Clinical data integration	Imaging, laboratories, vitals, symptoms, notes, treatment events
AI insights	Structured, evidence-linked, provenance-aware
Evaluation	Defined quantitative and qualitative metrics
Final deliverable	Running application, source code, test results, documentation, and demonstration

The cohort counts are engineering targets, not a requirement to process all data before the first functional release.

A smaller validated dataset is preferable to a larger inconsistent dataset.

3. Product Scope

3.1 In-scope clinical domain

The initial prototype shall support breast cancer.

Relevant characteristics may include:

- Patient age at baseline.
- Breast side.
- Confirmed cancer diagnosis.
- Tumor subtype.
- Tumor grade.
- Stage, where available.
- ER, PR, and HER2 status.
- Primary lesion location.
- Tumor dimensions and volume.
- Recorded treatment history.
- Available follow-up measurements.
- Clinical symptoms and observations.

Unknown values shall remain explicitly unknown.

The system must not infer receptor status, grade, or cancer stage solely from routine blood tests or MRI.

3.2 In-scope data modalities

Medical imaging

Primary modality:

- Breast MRI.

Optional modality:

- Breast-region CT where relevant, suitable data are available and modality-specific processing has been validated.

Supported information includes:

- Imaging study identifier.
- Acquisition date.
- Imaging series.
- Pixel spacing and orientation.
- Slice spacing or thickness.
- Image dimensions.
- Segmentation mask.
- Tumor volume and dimensions.
- Source and provenance information.

Laboratory results

Initially supported laboratory information may include:

- Hemoglobin.
- White blood cell count.
- Absolute neutrophil count when available.
- Platelet count.
- Selected liver-function measurements.
- Selected kidney-function measurements.
- Other explicitly configured and clinically relevant observations.

Every measurement must include its value, unit, collection date, provenance, and reference interval where a suitable source is available.

Vital signs

- Blood pressure.
- Heart rate.
- Body temperature.
- Weight.

- Additional explicitly configured measurements where available.

Symptoms

- Pain.
- Fatigue.
- Swelling.
- Nausea.
- Other defined research symptoms.

Clinical observations

- Physician examination notes.
- Imaging assessment summaries.
- Documented adverse effects.
- Recorded response assessments.
- Clinical concerns.
- Follow-up notes.

Treatment events

- Treatment category.
- Treatment regimen where documented.
- Start and end dates.
- Relevant treatment-event information.
- Changes or interruptions in the recorded treatment timeline.

Treatment administration and prescribing are out of scope.

3.3 Imaging scope

The prototype shall visualize the relevant breast region and segmented lesion.

It shall not require full-body reconstruction.

The system may display simplified surrounding breast anatomy to provide spatial context.

Such anatomy shall be identified as either derived from imaging or illustrative, depending on its source.

The digital twin shall not claim to visualize unmeasured biological properties such as drug resistance, hypoxia, necrosis, or immune infiltration as established patient findings without appropriate supporting observations.

3.4 Longitudinal scope

The prototype shall support three distinct representations:

1. **Observed longitudinal records:** Multiple actual visits or measurements.
2. **Synthetic longitudinal branches:** Artificial follow-up visits generated from a source baseline.
3. **Exploratory predicted states:** Future estimates generated by a documented model when supported.

These categories must never be displayed interchangeably.

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4. Target Users and Roles

4.1 Primary persona — Oncology researcher

The researcher needs to investigate tumor evolution, compare longitudinal measurements, explore hypothetical scenarios, and inspect supporting evidence.

Typical activities:

- Select a research case.
- Examine the anatomical model.
- Review laboratory observations.
- Compare treatment-associated trajectories.
- Inspect AI research insights.
- Evaluate uncertainty and provenance.

4.2 Secondary persona — Clinician reviewer

The clinician reviewer needs a concise representation of the available findings, temporal changes, and relevant evidence.

The prototype shall allow the reviewer to identify what has been observed, what is simulated, and what remains uncertain.

It shall not provide operational clinical treatment orders.

4.3 Technical persona — Research administrator

The research administrator prepares datasets, configures simulations, reviews validation results, and manages the technical demonstration.

Responsibilities may include:

- Importing dataset manifests.
- Selecting baseline samples.
- Running simulation batches.
- Reviewing generation failures.
- Configuring evaluation cases.
- Inspecting processing logs.

4.4 Access model

The prototype requires a lightweight access model rather than a complete hospital authorization infrastructure.

Minimum roles:

Role	Permissions
Research viewer	Read approved cases, timelines, and insights
Research editor	Add observations and run approved scenarios
Research administrator	Import research data and manage simulations

Access controls must protect source data and prevent unauthorized modifications.

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5. Core User Journeys

UJ-01 — Open an existing research patient

Actor: Researcher

Precondition: A suitable patient research profile exists.

Workflow:

1. Open the CancerOS research dashboard.
2. View the patient research case list.
3. Select a case by pseudonymous identifier.
4. Open the patient summary.
5. Review the baseline study date, available clinical information, and source data status.
6. Open the digital twin workspace.
7. Inspect the baseline tumor reconstruction and clinical measurements.

Expected outcome:

The researcher sees a coherent representation of the selected patient at baseline.

Acceptance criteria:

- Patient selection loads the correct profile.
- Only permitted source metadata are displayed.
- Tumor geometry corresponds to the associated case.
- Missing clinical information is shown as unavailable.
- No other patient's data are mixed into the current case.

UJ-02 — Import a real baseline case

Actor: Research administrator

Workflow:

1. Select the import function.
2. Provide an approved MRI study or dataset manifest.
3. Supply available clinical metadata.
4. Validate the source identifiers and imaging geometry.
5. Associate an existing segmentation mask or initiate a supported segmentation workflow.
6. Preview extracted measurements.

7. Confirm the import.

Expected outcome:

CancerOS creates a pseudonymous baseline patient record linked to its imaging and supporting data.

Acceptance criteria:

- Invalid or unsupported imaging is rejected with an understandable error.
- Valid imaging retains required spatial metadata.
- Source lineage is recorded.
- The application does not invent missing pathology or clinical values.
- A successfully imported case can be reopened and inspected.

UJ-03 — Generate synthetic patient evolution

Actor: Researcher or research administrator

Workflow:

1. Open a patient baseline.
2. Select the simulation workspace.
3. Choose an eligible scenario family.
4. Choose an appropriate simulated treatment-exposure category.
5. Specify permitted simulation settings or accept defaults.
6. Generate longitudinal observations.
7. Review validation status.
8. Open the generated branch in the timeline viewer.

Expected outcome:

The system produces a reproducible, time-indexed hypothetical clinical trajectory.

Acceptance criteria:

- The authentic baseline remains unchanged.
- Synthetic follow-up records are explicitly labeled.

- Generated visits follow chronological order.
- Treatment-event compatibility rules are checked.
- Laboratory values retain valid units.
- Critical data validation failures block publication of the branch.
- Repeating the same simulation version, seed, and parameters reproduces the same outputs.

UJ-04 — Compare tumor evolution

Actor: Researcher

Workflow:

1. Open a longitudinal branch.
2. Select the first observation.
3. Inspect tumor geometry and volume.
4. Move to a later observation.
5. Observe the updated tumor geometry.
6. Compare the numerical measurements.
7. Review associated clinical and treatment events.

Expected outcome:

The researcher can explore the progression of the patient state over time.

Acceptance criteria:

- The time selector changes the displayed state.
- The viewer identifies whether that state is observed or synthetic.
- Volume differences are calculated consistently.
- Clinical charts follow the selected visit.
- Switching dates does not overwrite previous data.

UJ-05 — Compare alternative synthetic trajectories

Actor: Researcher

Workflow:

1. Open a real baseline patient.
2. Select two or more generated branches.
3. Compare their tumor measurements.
4. Review corresponding treatment exposures.
5. Inspect laboratory, vital-sign, and symptom trends.
6. Review the simulation assumptions behind each branch.

Expected outcome:

The user can compare hypothetical disease trajectories sharing a common baseline.

Acceptance criteria:

- All selected branches refer to the same source baseline.
- Each branch retains separate identifiers and data histories.
- Differences are displayed as simulated scenario outcomes.
- The application does not rank treatment options as clinically superior.
- The comparison explains that simulated branches do not establish patient-specific treatment effects.

UJ-06 — Add a new clinical observation

Actor: Research editor

Workflow:

1. Open an existing research patient.
2. Select Add Observation.
3. Choose the observation date.
4. Enter supported clinical measurements or attach a compatible imaging study.
5. Submit the observation.
6. Run validation and measurement processing.
7. Refresh the digital twin state.

8. Review updated comparisons and AI insights.

Expected outcome:

A new timestamped observation is incorporated into the patient timeline.

Acceptance criteria:

- Existing observations are preserved.
- The new observation retains its collection date.
- Measurements with invalid units or values are rejected or flagged.
- New calculations are associated with the correct observation.
- The system identifies changes from preceding observations.
- AI outputs are refreshed or clearly marked as stale pending recomputation.

UJ-07 — Request AI interpretation

Actor: Researcher or clinician reviewer

Workflow:

1. Open a patient at a selected observation date.
2. Request an AI research interpretation.
3. CancerOS gathers relevant observed or synthetic records.
4. Deterministic analysis calculates changes and trends.
5. The knowledge service retrieves applicable evidence.
6. The AI generates a structured explanation.
7. The interface displays findings, evidence, limitations, and uncertainty.

Expected outcome:

The researcher receives a transparent explanation of the selected state and relevant changes.

Acceptance criteria:

- Quantitative measurements match their supporting records.
- AI explanations identify the source of each relevant finding.
- Synthetic data are explicitly described as simulated.

- Missing evidence is disclosed.
- The system does not invent test results, treatments, or patient outcomes.
- Unsupported diagnoses and treatment prescriptions are not presented.
- The user can inspect the referenced supporting observations.

UJ-08 — Inspect genuine longitudinal imaging

Actor: Researcher

Workflow:

1. Select a patient with multiple actual imaging sessions.
2. Open the chronological MRI studies.
3. Inspect study-specific segmentation results.
4. Compare tumor measurements across visits.
5. Review differences and measurement uncertainty.

Expected outcome:

CancerOS demonstrates actual longitudinal imaging analysis separately from its synthetic simulation capabilities.

Acceptance criteria:

- Each imaging date corresponds to an authentic study.
- Real images are not replaced by generated approximations.
- Computed measurements retain their method and provenance.
- Temporal changes are displayed without claiming an unvalidated biological mechanism.

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6. Functional Requirements

6.1 Patient and research case management

FR-001: The system shall create and retrieve pseudonymous patient research profiles.

FR-002: Each patient shall have a stable internal identifier.

FR-003: Each patient may have multiple observation visits.

FR-004: Every observation shall include an observation date and provenance classification.

FR-005: The system shall distinguish absent values from normal or zero values.

FR-006: The system shall support case selection, inspection, and permitted updates.

FR-007: Source records shall remain auditable and shall not be silently overwritten.

6.2 Medical imaging

FR-008: The system shall accept approved breast MRI input formats defined in the imaging specification.

FR-009: The system shall preserve required spatial imaging metadata.

FR-010: The system shall support imported segmentation masks.

FR-011: Where a suitable model is available, the system may generate an automated segmentation proposal.

FR-012: The system shall calculate lesion measurements using documented physical spacing and geometry.

FR-013: The system shall identify the segmentation method and uncertainty where available.

FR-014: The system shall render the segmented lesion in an interactive 3D viewer.

FR-015: The viewer shall support rotation, zoom, and selection of available time states.

FR-016: Any illustrative surrounding anatomical structures shall be distinguished from measured anatomy.

6.3 Synthetic longitudinal simulation

FR-017: The system shall create synthetic branches linked to an authentic baseline case.

FR-018: The generator shall support five primary clinical trajectory families.

FR-019: Each scenario shall support at least four dated observation states in the curated demonstration.

FR-020: Simulation shall account for treatment-exposure categories and defined compatibility constraints.

FR-021: Tumor response and patient clinical deterioration shall be modeled as separable dimensions.

FR-022: The simulator shall generate supported clinical observations with valid measurement units and timestamps.

FR-023: Synthetic tumor geometry shall remain distinguishable from actual medical imaging.

FR-024: The generator shall use recorded simulation parameters and a random seed.

FR-025: Generated records shall pass configured clinical and structural consistency checks before publication.

FR-026: The original patient baseline shall remain unmodified.

FR-027: The system shall allow multiple independent hypothetical branches from the same baseline.

FR-028: Every generated observation shall retain its simulation configuration and parent case identifier.

6.4 Longitudinal evolution and temporal analysis

FR-029: The system shall maintain time-ordered patient state histories.

FR-030: The system shall calculate changes in supported measurements between visits.

FR-031: The system shall show tumor volume changes alongside available laboratory and clinical measurements.

FR-032: The system shall align recorded treatment events with the patient timeline.

FR-033: The system shall distinguish observed change, interpolation, and model prediction.

FR-034: The system shall preserve historical states when new records are added.

FR-035: Where a forecasting method is enabled, the system shall show prediction uncertainty and a stated prediction horizon.

FR-036: Forecast generation shall be declined or qualified when sufficient information is not available.

6.5 Multimodal AI interpretation

FR-037: CancerOS shall construct a structured context from relevant patient observations.

FR-038: Deterministic calculations shall be used for measurable numerical changes.

FR-039: AI-generated explanations shall be grounded in the assembled context and retrieved evidence.

FR-040: The system shall distinguish source observations from interpretations and hypotheses.

FR-041: The interpretation service shall identify missing or contradictory evidence where relevant.

FR-042: Generated insights shall reference supporting records or research evidence.

FR-043: The system shall avoid unsupported treatment recommendations, diagnoses, and biological mechanism claims.

FR-044: The system shall retain the model version and generation metadata required for auditing.

FR-045: If the AI service is unavailable, source measurements and visualizations shall remain accessible.

6.6 Clinical knowledge and historical evidence

FR-046: The system shall support a small, explicit clinical knowledge representation.

FR-047: The knowledge layer shall relate supported patient, tumor, treatment, observation, and outcome entities.

FR-048: Each evidence relationship shall carry its provenance and evidence category.

FR-049: The system shall distinguish synthetic associations from those supported by real patient records or published research.

FR-050: Historical retrieval shall not imply that comparable patients necessarily share treatment responses.

6.7 User interface

FR-051: The interface shall provide a patient case selection workspace.

FR-052: The interface shall provide an anatomical digital twin workspace.

FR-053: The interface shall provide a longitudinal timeline and observation selector.

FR-054: The interface shall provide synchronized clinical measurement charts.

FR-055: The interface shall provide a synthetic scenario generation and comparison workflow.

FR-056: The interface shall provide a structured AI insight panel.

FR-057: The interface shall visibly label source classifications.

FR-058: Errors, empty states, and unavailable capabilities shall be communicated clearly.

FR-059: Core research workflows shall use understandable medical and research terminology rather than unexplained engineering abbreviations.

6.8 Evaluation and demonstration

FR-060: The system shall provide reproducible demonstration cases.

FR-061: The engineering team shall maintain held-out evaluation data partitioned by original patient identity.

FR-062: Imaging evaluation shall report applicable segmentation and measurement metrics.

FR-063: Temporal evaluation shall distinguish genuine observed outcomes from synthetic simulation outcomes.

FR-064: The project shall document the limitations of its datasets and models.

FR-065: The complete demonstration shall execute without modifying database records manually.

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7. Synthetic Clinical Scenario Requirements

Five scenario families are required.

SC-01 — Strong response

The generated trajectory shall demonstrate a sustained reduction in the selected tumor measurement over the simulated observation period.

The system may also generate compatible changes in symptoms and other clinical indicators.

It shall not assume all laboratory values improve automatically with tumor response.

SC-02 — Partial response or stable disease

The generated trajectory shall demonstrate modest change or relative stability in tumor measurements.

The scenario must account for measurement uncertainty and shall not automatically interpret small changes as meaningful clinical progression.

SC-03 — Progressive disease or poor response

The generated trajectory shall demonstrate increasing tumor burden despite a recorded hypothetical treatment exposure.

The AI shall identify the measured or simulated increase.

It shall not claim a confirmed molecular resistance mechanism based solely on increasing tumor size.

SC-04 — Initial response followed by regrowth

The generated trajectory shall demonstrate an initial reduction followed by a later increase.

The system shall retain each intermediate state so the user can inspect the change in direction.

The model may flag the pattern for investigation but shall not assert the cause.

SC-05 — Clinical deterioration or treatment toxicity

The generated trajectory shall demonstrate worsening selected clinical indicators that may occur independently of tumor response.

Examples may include selected blood-count abnormalities, adverse symptoms, or worsening vital signs.

The application must avoid implying that these simulated findings establish toxicity causation in a real patient.

7.1 Treatment-exposure categories

The prototype shall represent five broad categories:

1. Chemotherapy.
2. Endocrine therapy.
3. HER2-targeted therapy.
4. Immunotherapy-containing treatment.
5. Surgery and/or radiotherapy pathways.

Treatment categories shall be restricted by explicit research-case compatibility rules.

Sequential or combined treatment events may be represented when supported by the simulation configuration.

The prototype shall not rank these categories as clinically optimal for an individual patient.

7.2 Scenario configuration

Each generated branch shall retain:

- Source baseline identifier.
- Scenario family.
- Treatment-exposure configuration.
- Simulated observation dates.
- Generation model or rule version.
- Random seed.
- Source assumptions.
- Validation outcome.
- Generation timestamp.
- Explicit synthetic-data classification.

7.3 Data consistency rules

Synthetic records must satisfy at least the following:

- Valid chronological ordering.
- Standardized and interpretable measurement units.
- No impossible or invalid field combinations under the defined schema.
- No treatment exposure requiring a known biological status without an appropriate compatibility rule.
- No fabricated clinical fact presented as a source observation.
- No transformation of generated image geometry into purported authentic imaging evidence.
- No mixing of records between unrelated source patients.
- No use of synthetic outputs as independent evidence of actual treatment efficacy.

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8. Data Requirements and Source Strategy

8.1 Primary source principle

Real, properly de-identified medical imaging and available clinical data shall form the baseline of the research cases.

Artificial observations shall supplement these cases for workflow demonstration and controlled experimentation.

The prototype shall not claim that synthetic records constitute genuine longitudinal patient outcomes.

8.2 Candidate datasets

Candidate sources include:

- I-SPY2 longitudinal breast MRI.
- I-SPY1 longitudinal breast MRI.
- Duke Breast Cancer MRI.
- BreastDCEDL processed imaging resources.

Dataset selection shall depend on access permissions, available imaging, relevant metadata, segmentation quality, longitudinal coverage, and license conditions.

The final dataset specification must verify exact variables before they are treated as supported fields.

8.3 Dataset independence

Overlapping source collections must not be counted as independent patients.

Data partitions shall be assigned using original patient identity before synthetic trajectories are generated.

Every branch descended from one source patient must remain within that patient's assigned partition.

8.4 Synthetic data policy

The system shall maintain the following provenance classifications:

- REAL_OBSERVED
- REAL_DERIVED
- SYNTHETIC_GENERATED
- INTERPOLATED
- MODEL_PREDICTED

Source records, synthetic observations, and AI results must remain distinguishable at the database, service, and user-interface levels.

8.5 Missing information

The prototype shall handle incomplete records without inventing absent values.

Missing data must not be silently interpreted as normal.

Optional imputation or synthetic supplementation must be explicitly recorded and labeled.

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9. Digital Twin Product Requirements

9.1 Twin composition

Each digital twin shall contain:

1. A persistent patient research identity.
2. Available source imaging and segmentation geometry.
3. Time-indexed tumor measurements.
4. Clinical observations associated with actual or simulated dates.
5. Recorded treatment events.
6. Provenance information.
7. Derived longitudinal comparisons.
8. Structured research interpretations and uncertainty.

9.2 Twin state model

A patient twin may contain multiple time states.

Each state shall identify:

- The relevant observation timestamp.
- Available imaging or associated geometry.
- Tumor measurements.
- Laboratory observations.
- Vital signs.

- Symptoms.
- Treatment-event context.
- Data source classification.
- Derived insights.

Where information is not available for a state, the corresponding value must remain missing.

9.3 Anatomical visualization

The user shall be able to:

- Rotate the tumor model.
- Zoom in and out.
- View available lesion boundaries.
- Switch between visits.
- View associated measurements.
- Compare earlier and later states.

The viewer may simplify anatomical surroundings to keep rendering efficient.

9.4 Longitudinal visualization

The interface shall include a timeline representing available visits.

Selecting a visit shall synchronize the anatomical representation with the corresponding clinical records.

Observed and synthetic states shall be visually differentiated.

Interpolation between visits, if implemented, shall be clearly marked as an estimate.

9.5 Scenario branching

The same baseline case may have multiple hypothetical futures.

These branches shall be independent of one another after the common baseline.

Generating or updating one branch must not modify other branches.

A synthetic branch must never replace a patient's actual historical record.

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10. AI Interpretation and Safety Requirements

10.1 Permitted AI outputs

The prototype may generate:

- Summaries of available findings.
- Changes in recorded tumor measurements.
- Descriptions of laboratory trends.
- Descriptions of symptom changes.
- Temporal comparison summaries.
- Relevant uncertainty statements.
- Missing-data observations.
- References to supporting clinical information.
- Research hypotheses explicitly labeled as such.

10.2 Prohibited or restricted outputs

The prototype shall not present:

- Autonomous clinical diagnoses.
- Patient-specific treatment prescriptions.
- Claims of clinically validated treatment optimization.
- Unsupported survival probabilities.
- Unsupported molecular resistance mechanisms.
- Synthetic findings as actual medical outcomes.
- Fictional measurements as genuine test results.
- Simulated comparisons as causal estimates of treatment effectiveness.

10.3 Evidence requirements

AI outputs shall identify the records and evidence used.

Numerical claims shall originate from validated data or deterministic calculations rather than unrestricted language-model generation.

If a relevant record is absent, the AI shall disclose the limitation.

10.4 Model uncertainty

The system shall disclose:

- When data are incomplete.
- When images or measurements are synthetic.
- When estimates are interpolated.
- When a forecast has insufficient support.
- When an interpretation is exploratory rather than clinically validated.

10.5 Medical disclaimer

All research interfaces shall display an appropriate statement establishing that the prototype is not validated for medical diagnosis or treatment decisions.

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11. User Interface and Experience Requirements

11.1 Required screens

Screen A — Research case dashboard

Features:

- Available research cases.
- Pseudonymous identifiers.
- Cancer subtype information where available.
- Number of observations.
- Provenance summary.
- Access to the digital twin workspace.

Screen B — Patient research profile

Features:

- Baseline clinical summary.
- Tumor information.

- Available imaging sessions.
- Clinical record completeness.
- Recorded treatments.
- Access to available simulation branches.

Screen C — Digital twin workspace

Features:

- Interactive 3D tumor viewer.
- Date selector.
- Tumor measurements.
- Timeline.
- Laboratory and symptom charts.
- Data provenance indicators.

Screen D — Simulation workspace

Features:

- Baseline selection.
- Scenario-family selection.
- Treatment-exposure selection.
- Generation controls.
- Simulation validation results.
- Branch navigation.

Screen E — AI research insights

Features:

- Structured observations.
- Temporal comparisons.
- Evidence references.
- Uncertainty statements.

- Clear identification of simulated and observed results.

Screen F — Evaluation workspace

Features:

- Available test cases.
- Evaluation metrics.
- Model or simulation version information.
- Data source descriptions.
- Known limitations.

The evaluation workspace may be a compact read-only screen for the hackathon.

11.2 Design principles

The interface shall prioritize:

- Scientific clarity.
- Visual coherence.
- Fast navigation.
- Clear state changes.
- Legible clinical measurements.
- Low cognitive load.
- Consistent terminology.
- Transparent uncertainty.

A highly polished interface is desirable, but visual effects must not misrepresent the medical data.

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12. Nonfunctional Requirements

12.1 Performance

NFR-001: The application shall provide responsive navigation for the curated demonstration cases.

NFR-002: The 3D viewer shall remain usable during normal rotation and time-state selection on the selected demonstration environment.

NFR-003: Expensive imaging preprocessing should occur asynchronously or ahead of the demonstration, rather than blocking routine interface navigation.

NFR-004: AI inference delays shall be communicated through visible processing states.

12.2 Reliability

NFR-005: A failed AI request shall not corrupt clinical or imaging records.

NFR-006: Failed simulation generation shall not publish incomplete branches.

NFR-007: Invalid input shall produce clear validation feedback.

NFR-008: Precomputed demonstration assets shall remain available if optional AI generation is temporarily unavailable.

12.3 Security and privacy

NFR-009: Source patient records shall be pseudonymous.

NFR-010: Protected source data shall not be submitted to external AI services without appropriate authorization and policy verification.

NFR-011: Secrets and credentials shall not be embedded in frontend code.

NFR-012: Research resources shall require appropriate access control.

NFR-013: Dataset access and use shall respect the relevant licenses and restrictions.

12.4 Maintainability

NFR-014: The application shall separate research data management, imaging, simulation, AI interpretation, and visualization responsibilities.

NFR-015: Model and simulation services shall expose documented interfaces.

NFR-016: The implementation shall include reproducible environment configuration and clear startup instructions.

NFR-017: Core calculation, provenance, and simulation modules shall have automated tests.

12.5 Hardware and deployment

Development shall accommodate a local Windows laptop with limited memory and no high-capacity dedicated machine-learning GPU.

The prototype should support:

- Lightweight local application development.
- Preprocessed or downsampled demonstration assets.
- Remote inference or GPU execution where necessary.
- Efficient loading of imaging-derived 3D geometry.
- Deployment to a publicly accessible or judge-accessible demonstration environment, subject to competition rules.

Large-scale local medical-imaging model training shall not be a prerequisite.

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13. Explicit Exclusions

The following features shall not be implemented in the hackathon MVP unless the project scope is formally revised.

Excluded capability	**Reason**
Other cancer types	Focus on one coherent clinical domain
Whole-body digital twin	Unnecessary anatomical and engineering scope
Full physiological simulation	Insufficient modeling time and validation data
Full genomic sequencing analysis	Outside selected minimum dataset
Drug discovery	Separate long-term product capability
Autonomous therapy selection	Clinical validation and safety requirements
Prescription management	Hospital operations functionality
Treatment administration	Outside research prototype scope
Full hospital records platform	Unnecessary platform complexity
Wearable monitoring ecosystem	Outside selected modalities
Endoscopy and surgical video	Outside breast MRI-focused scope
Real-time hospital integrations	Access and implementation complexity

Reinforcement learning oncology agent	Inappropriate for the prototype's evidence base
Global clinical knowledge network	Outside hackathon scale
Clinically validated survival prediction	Insufficient data and independent validation
Production medical-device certification	Outside research prototype scope
Automated continuous model retraining	Unnecessary and difficult to validate within hackathon constraints

These exclusions are deliberate product boundaries, not limitations of the long-term CancerOS vision.

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14. Feature Prioritization

P0 — Mandatory for hackathon readiness

- Real MRI baseline ingestion or curated dataset loading.
- Correct research patient identity and observation management.
- Working tumor segmentation or validated imported masks.
- Functional 3D tumor visualization.
- Five synthetic scenario families.
- Reproducible longitudinal record generation.
- Four-visit scenario timeline.
- Laboratory, vital-sign, symptom, and treatment-event display.
- Provenance-aware observed/synthetic state separation.
- Patient-state updates.
- AI-assisted structured research interpretations.
- Basic evidence references and uncertainty statements.
- At least one genuine longitudinal imaging demonstration.

- Automated integrity tests.
- Functional evaluation and limitations documentation.

P1 — High-value enhancements

- Interactive comparison of multiple scenario branches.
- More sophisticated temporal prediction methods.
- Expanded clinical evidence retrieval.
- Additional segmentation model comparisons.
- Enhanced anatomical visualization.
- Research summary export.
- More extensive dataset processing.

P2 — Post-hackathon

- Larger research cohorts.
- Expanded clinical integrations.
- Additional imaging modalities.
- More advanced mechanistic models.
- Prospective observational validation.
- Additional cancer types.
- Hospital and research-institution integrations.

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15. Acceptance Criteria and Release Gates

The prototype shall not be considered ready solely because its dashboard looks complete.

Release readiness requires successful demonstration of the following capabilities.

AC-01 — Source data integrity

Given a valid approved research MRI case,
When the system imports or loads it,
Then the patient identity, imaging metadata, source information, and available annotations are correctly preserved.

Pass condition: No critical patient-source mismatches in the curated test suite.

AC-02 — Anatomical visualization

Given a valid imaging-derived segmentation,
When the patient is opened,
Then CancerOS renders a navigable 3D tumor representation.

Pass condition: All designated demonstration cases load and display their correct lesion geometry.

AC-03 — Longitudinal simulation

Given an eligible baseline and scenario configuration,
When the simulation engine generates follow-up states,
Then the system produces chronologically consistent observations linked to the correct original patient.

Pass condition: All five primary scenario families run successfully in the curated test suite.

AC-04 — Reproducibility

Given the same baseline, simulator version, parameters, and random seed,
When the generation process is repeated,
Then the same intended synthetic outputs are reproduced.

Pass condition: Repeated deterministic tests produce matching records within documented numerical tolerances.

AC-05 — Provenance separation

Given a patient state containing authentic and synthetic observations,
When the researcher inspects the timeline or AI findings,
Then each observation is classified correctly.

Pass condition: No mislabeled synthetic measurements in the curated test suite.

AC-06 — Timeline synchronization

Given a patient with multiple observations,
When a different date is selected,
Then the tumor geometry, measurement summaries, charts, and clinical events update to the appropriate state.

Pass condition: The selected state and displayed records match across all integrated components.

AC-07 — Historical immutability

Given an existing patient timeline,
When a new observation is added,
Then the previous states remain unchanged.

Pass condition: Automated tests confirm preservation of prior observations and correct insertion of the new state.

AC-08 — AI interpretation grounding

Given a patient with available evidence,
When an AI interpretation is requested,
Then the result uses supported measurements and identifies meaningful limitations.

Pass condition: No invented quantitative findings in the curated AI evaluation set.

AC-09 — Missing-data handling

Given a patient with incomplete clinical observations,
When the system generates an analysis,
Then missing values are reported rather than fabricated.

Pass condition: All defined missing-data test cases produce appropriate results.

AC-10 — Real longitudinal evaluation

Given a suitable independent set of authentic follow-up MRI studies,
When tumor measurements or temporal estimates are evaluated,
Then the system reports appropriate quantitative metrics and limitations.

Pass condition: A reproducible evaluation report exists, including sample size, methodology, results, and comparison against a simple baseline where forecasting is attempted.

No minimum clinical accuracy claim is established at the PRD stage.

AC-11 — Scientific safety

Given a request that would require unsupported clinical conclusions,
When the AI produces an explanation,
Then it clearly limits its conclusions to the available research evidence.

Pass condition: Defined safety tests produce no unsupported patient-specific therapy prescriptions or fabricated diagnostic conclusions.

AC-12 — Complete hackathon demonstration

Given a prepared deployment and approved demonstration dataset,
When the full demonstration is executed,
Then the presenter can load a patient, visualize the tumor, explore a synthetic trajectory, add or process an observation, and inspect an AI interpretation.

Pass condition: The complete integrated demonstration succeeds without manual modification of internal application data.

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16. Key Dependencies and Risks

16.1 Longitudinal clinical data availability

Risk: Real datasets may not provide matching follow-up imaging, laboratory measurements, treatment histories, and symptoms.

Mitigation: Use real data where available and explicitly synthetic observations where necessary. Reserve genuine repeated-imaging cases for independent evaluation.

16.2 Imaging model feasibility

Risk: A selected AI model may not reliably segment breast tumors.

Mitigation: Support verified masks, dedicated medical-imaging tools, and clearly documented segmentation methods.

16.3 Synthetic data validity

Risk: The generator may produce medically inconsistent combinations or unrealistic temporal patterns.

Mitigation: Use constrained simulation, validation rules, documented assumptions, and expert review where feasible.

16.4 False predictive confidence

Risk: Evaluation on simulator-generated data may create misleadingly strong performance metrics.

Mitigation: Separate synthetic evaluation from genuinely observed patient outcomes and compare temporal models with simple baselines.

16.5 AI hallucinations

Risk: The language model may invent measurements or clinical conclusions.

Mitigation: Use deterministic computation, constrained evidence contexts, structured output checks, and explicit limitations.

16.6 Time and compute constraints

Risk: Large medical imaging pipelines and multiple AI models could exceed available hardware or development capacity.

Mitigation: Use a modular architecture, processed demonstration assets, remote inference where justified, and P0-first delivery.

16.7 Data privacy and licensing

Risk: Research data may have restrictions that prevent particular uses or external processing.

Mitigation: Maintain a dataset license inventory, protect source information, and verify processing rights before transmitting data to external services.

16.8 Competition compliance

Risk: Project scope, submitted artifacts, or implementation timing may conflict with official competition rules.

Mitigation: Verify the published competition terms before implementation and final submission.

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17. Engineering Deliverables

Development shall produce:

1. Running CancerOS web application.
2. Backend service and documented APIs.
3. Pseudonymous clinical research database.
4. Approved MRI ingestion and imaging processing pipeline.
5. Breast-region tumor visualization.
6. Synthetic longitudinal research engine.
7. Five configured scenario families.
8. Temporal state and measurement comparison capability.
9. AI-assisted interpretation and evidence retrieval.

10. Provenance and scientific safety controls.
11. Automated testing and evaluation assets.
12. Reproducible demonstration cases.
13. Deployment configuration and setup instructions.
14. Final technical report documenting methods, results, and limitations.
15. Hackathon presentation materials and demonstration instructions.

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18. Implementation Governance

The Product Requirements Document is the primary definition of the prototype's intended product behavior.

Subsequent technical documents shall refine implementation details without expanding product scope unless a scope change is explicitly approved.

The following engineering specifications shall derive from this PRD:

- COS-SPEC-002 — Competition and Evaluation Strategy.
- COS-SPEC-003 — Clinical Data and Dataset Specification.
- COS-SPEC-004 — Synthetic Longitudinal Simulation Specification.
- COS-SPEC-005 — Medical Imaging and Digital Twin Specification.
- COS-SPEC-006 — Multimodal AI Specification.
- COS-SPEC-007 — Software and Infrastructure Architecture.
- COS-SPEC-008 — Database Schema and API Contracts.
- COS-SPEC-009 — UI/UX Specification.
- COS-SPEC-010 — AI Safety, Privacy and Scientific Integrity.
- COS-SPEC-011 — Testing and Evaluation Specification.
- COS-SPEC-012 — Implementation Roadmap and Finalist Demo Plan.

Each downstream document shall identify the PRD requirements it implements.

Coding agents must not replace source clinical information with invented values, reinterpret research simulations as validated clinical forecasts, or implement excluded capabilities without authorization.

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19. Final Definition of Product Success

CancerOS will meet its hackathon product objective when a researcher can:

1. Open a genuine breast cancer baseline case.
2. Inspect the authentic imaging-derived tumor.
3. Generate and select clearly labeled hypothetical disease trajectories.
4. Observe corresponding tumor and clinical changes over time.
5. Compare hypothetical trajectories without treating them as actual treatment outcomes.
6. Add a new research observation.
7. See the digital twin update without losing historical records.
8. Receive an evidence-linked AI interpretation.
9. Inspect data sources, uncertainty, and model limitations.
10. View quantitative results from appropriate real-data evaluation.

The defining achievement is a working transformation of isolated cancer observations into an interactive, evolving, multimodal research representation.

The project succeeds as a proof of concept by demonstrating that this transformation is technically coherent, reproducible, transparent, and useful for structured investigation.

It does not need to prove clinical treatment effectiveness to satisfy its prototype objective.

End of Document — COS-PRD-001, Version 1.0